The leading constant of the memory–sample tradeoff for noiseless Gaussian regression

[AUTHOR NAME]

Draft, October 2026

Abstract

A streaming learner reads exact Gaussian linear equations $Y_t = \langle X_t, s\rangle$, $X_t \sim N(0, I_d)$, keeps at most M bits between samples, and must output a unit vector at angular distance at most ϵ from s. A recent preprint shows that $M = o(d^2)$ forces $T \geq 2^{-16} d \log_2(1/\epsilon)$ samples. We show that the same conclusion holds with 2^{-16} replaced by any constant $c < 2\ln 2 \approx 1.386$, uniformly in $0 < \epsilon \leq 1/10$ (in fact with $\log_2(1/\epsilon)$ replaced by the slightly larger $\log_2(1/\sin\epsilon)$). We also prove that randomized Kaczmarz with fixed-point registers of $\log_2(1/\epsilon) + \frac{3}{2}\log_2 d + \log_2(1/\eta) + O(1)$ bits per coordinate succeeds after $(1+\eta)\, d \ln(1/\sin^2\epsilon) \approx (1+\eta)\, 2\ln 2 \cdot d \log_2(1/\epsilon)$ samples, for every signal. Hence for memory in the window $d(\log_2(1/\epsilon) + 2\log_2 d) \leq M = o(d^2)$ the optimal number of samples is $(2\ln 2 + o(1))\, d \log_2(1/\epsilon)$, and randomized Kaczmarz is first-order optimal. The improvement over the original proof comes from three local changes: a moving split point in its beta-mixture comparison, an exact beta-ratio estimate, and an arcsin form of the terminal cap bound. We also explain why the original fixed split at $1/2$ cannot give a constant above 0.5176, or above 0.3336 with the original ratio estimates. We record how slowly the bound converges at finite d.

1 Introduction

1.1 Model and previous bounds

We use the model of the OpenAI preprints [1, 2].

Definition 1 (Finite-state Gaussian stream [1, Definition 1.1])**.** Fix integers $d \geq 2$, $M \geq 0$, $T \geq 0$ and $0 < \epsilon \leq 1/10$. For a signal $s \in S^{d-1}$ the observations are $X_t \sim N(0, I_d)$ independent and $Y_t = \langle X_t, s\rangle$, $1 \leq t \leq T$. The learner reads the pairs in order. After each sample its persistent state takes at most 2^M values. Transitions may depend on the index, on d, M, T, ϵ, on the current state, on the *entire current real pair* (X_t, Y_t), and on fresh randomness. Computation inside a transition is unrestricted. All information retained for later samples must be in the state. The learner stops by time T and outputs $\widehat{s} \in S^{d-1}$ as a function of the terminal state, the stopping index and fresh randomness. Success means $\arccos\langle \widehat{s}, s\rangle \leq \epsilon$. Measurability conventions are as in [1].

Throughout, $\lambda = \log_2(1/\epsilon) \geq \log_2 10$. The model charges bits of *persistent* state. It does not charge real registers used inside one transition. In particular a transition may read the real pair exactly, combine it with the stored state in exact arithmetic, and store a rounded result. This is how the upper bound in Section 4 uses the model.

The preprint [1] proves: if $M(d) = o(d^2)$, then for all large d, every $0 < \epsilon \leq 1/10$ and every learner whose success probability under a uniform signal is at least $2/3$, $T \geq 2^{-16} d \log_2(1/\epsilon)$ [1, Theorem 1.2]. On the other side, randomized Kaczmarz [6, 7] with real registers satisfies

$\mathbb{E}\|z_t - s\|^2 = (1 - 1/d)^t$ [2, §1.2, eq. (2); source `introduction.tex`:201–227]. This suggests the scale $d\ln(1/\epsilon^2) = 2\ln 2 \cdot d\lambda$. However, the update stores real numbers, and [2] states explicitly that it "supplies no finite-bit upper bound". Sharan, Sidford and Valiant remark without proof that randomized Kaczmarz "can be easily shown" to reach error ϵ with $O(d\log(1/\epsilon))$ samples and $O(d\log(1/\epsilon))$ memory [3, §1.1]. Thus the order $d\log(1/\epsilon)$ was known for $o(d^2)$ memory up to a factor of about $2^{16} \cdot 2\ln 2$. Lemma 3 is a written-out version of that remark with explicit constants; we claim no novelty for it beyond the constants.

For context: with noisy labels and $d^2/4$ bits, Sharan, Sidford and Valiant [3] proved an $\Omega(d\log r)$ lower bound at accuracy d^{-r}, and left open whether the $O(d\log(1/\epsilon))$ samples of randomized Kaczmarz are optimal [3, §1.1]. The order question in their setting is addressed in [2], not here. Related memory lower bounds for optimization are in [4, 5]. The present note does *not* answer the question of [3]. It sharpens $\Theta(d\log 1/\epsilon)$ to the leading constant $2\ln 2$ for $o(d^2)$ memory. In this regime randomized Kaczmarz is first-order optimal.

1.2 Results

Theorem 2 (Lower bound with the sharp constant)**.** *Let $M : \mathbb{N} \to \mathbb{Z}_{\geq 0}$ satisfy $M(d) = o(d^2)$, and let $c < 2\ln 2$. There is $d_0 = d_0(c, M(\cdot))$ such that for every $d \geq d_0$, every $0 < \epsilon \leq 1/10$ and every learner of Definition 1 with memory $M(d)$, if S is uniform on S^{d-1} independently of the rows and the learner's randomness and $\mathbb{P}\{\arccos\langle\widehat{s}, S\rangle \leq \epsilon\} \geq 2/3$, then*

$$T \;\geq\; c\,d\log_2\frac{1}{\sin\epsilon} \;\geq\; c\,d\log_2\frac{1}{\epsilon}.$$

Lemma 3 (Finite-bit randomized Kaczmarz)**.** *Let $0 < \eta \leq 1$, $d \geq 16/\eta^2$ and $0 < \epsilon \leq 1/10$. Put*

$$T = \big\lceil (1+\eta)\,d\ln(1/\sin^2\epsilon)\big\rceil, \qquad q = \Big\lceil \log_2\frac{1}{\sin\epsilon} + \frac{3}{2}\log_2 d + \log_2\frac{4\sqrt{6}}{\eta}\Big\rceil.$$

There is a learner of Definition 1 with

$$M = d(q+3) \;\leq\; d\Big(\lambda + \frac{3}{2}\log_2 d + \log_2\frac{1}{\eta} + 7.3\Big)$$

bits, horizon T, and a deterministic transition rule (fixed-point randomized Kaczmarz). For every *signal $s \in S^{d-1}$ it satisfies $\mathbb{P}_s\{\arccos\langle\widehat{s}, s\rangle \leq \epsilon\} \geq 2/3$. Moreover*

$$d\ln\frac{1}{\sin^2\epsilon} \;\leq\; d\ln\frac{1}{\epsilon^2} + \frac{d\epsilon^2}{2} \;\leq\; 1.0011 \cdot 2\ln 2 \cdot d\lambda.$$

Hence $T \leq (1+\eta)\,1.0011 \cdot 2\ln 2 \cdot d\lambda + 1$.

In the form requested by the lower bound, Lemma 3 gives $M \leq d(\lambda + C\log_2 d)$ with $C = \frac{3}{2} + (\log_2(1/\eta) + 7.3)/\log_2 d$. For example, $C = 2$ works once $d \geq 2^{14.6}\eta^{-2}$. The range $0 < \epsilon \leq 1/10$ is uniform, and $d_0 = 16/\eta^2$ does not depend on ϵ.

Corollary 4 (Sharpness window)**.** *Let $M(d) = o(d^2)$, and let $\epsilon = \epsilon_d \in (0, 1/10]$ satisfy $d(\lambda_d + 2\log_2 d) \leq M(d)$ for all large d. Let $T^*(d)$ be the least horizon for which some learner with memory $M(d)$ reaches accuracy ϵ_d with probability $\geq 2/3$ for every signal. (The same conclusions hold for the least horizon with success probability $\geq 2/3$ under a uniform signal, which lies between the two bounds below.) Then*

$$\lim_{d\to\infty} \frac{T^*(d)}{d\ln(1/\sin^2\epsilon_d)} = 1,$$

hence

$$2\ln 2 \le \liminf_d \frac{T^*(d)}{d\log_2(1/\epsilon_d)} \le \limsup_d \frac{T^*(d)}{d\log_2(1/\epsilon_d)} \le 1.0011 \cdot 2\ln 2,$$

with $\lim T^*(d)/(d\log_2(1/\epsilon_d)) = 2\ln 2$ *whenever* $\epsilon_d \to 0$.

Proof. Note $d\ln(1/\sin^2\epsilon) = 2\ln 2 \cdot d\log_2(1/\sin\epsilon)$. For the lower bound, an every-signal guarantee implies the uniform-prior guarantee [1, Corollary 1.3], so Theorem 2 gives $\liminf T^*/(d\ln(1/\sin^2\epsilon_d)) \ge c/(2\ln 2)$ for every $c < 2\ln 2$. For the upper bound, fix $\eta \in (0,1]$. Once $\frac{1}{2}\log_2 d \ge \log_2(1/\eta) + 7.3$ and $d \ge 16/\eta^2$, Lemma 3 applies within memory $d(\lambda + 2\log_2 d) \le M(d)$, giving $T^* \le (1+\eta)d\ln(1/\sin^2\epsilon_d) + 1$. Let $\eta \downarrow 0$. The second display follows from $1 \le \ln(1/\sin^2\epsilon)/\ln(1/\epsilon^2) \le 1 + \epsilon^2/(2\ln(1/\epsilon^2)) \le 1.0011$ (Lemma 3). □

Remark 5 (Window and the $\Theta(d^2)$ regime). Corollary 4 needs both ends of the window. If $M < d\lambda$, the learner cannot store λ bits per coordinate, and our upper bound does not apply. If $M = Ad^2$ with fixed $A > 0$, Theorem 2 is not available. The proof below then yields only a smaller constant: $c(A) \approx 0.47, 0.72, 0.93, 1.19, 1.31$ for $A = 10^{-2}, 10^{-3}, 10^{-4}, 10^{-6}, 10^{-8}$ (script `computations/memratio.py`, asymptotic in d; heuristic, since it treats the dimension loss $2J$ as negligible, while for $M = \Theta(d^2)$ the short-run lemma only gives $\lambda = O(d)$). The sharp constant for $\Theta(d^2)$ memory is unknown.

1.3 What is new

The overall architecture of the lower bound is that of [1]: residual subspheres, an m-th moment to pay for state selection, a two-component beta mixture for correlated replica directions, backward induction over blocks, and a terminal cap. We change three estimates and re-optimize the parameters.

(i) *Moving split* (Lemma 7). The original two-component domination splits the beta density at $q_0 = 1/2$. This costs a factor $2^{(b-1)/2}$ per replica direction, which is $(\kappa/2)\ln 2$ nats per unit dimension per block, independently of all parameters. This is at least as large as the residual cost $(\kappa/2)\ln\frac{1}{1-\pi}$ whenever $\pi \le 1/2$, so no choice of parameters removes it. Splitting at $q_0 = 1-\delta$ with $\delta \to 0$ reduces this cost to $(\kappa/2)\ln\frac{1}{1-\delta} = O(\kappa\delta)$.

(ii) *Beta-ratio bound* (Lemma 8). The price of the moving split is a factor $\delta^{-(b-2)/2}$ on the second component. It is cancelled by the exact ratio $\mathrm{B}(\alpha,\beta_1)/\mathrm{B}(\alpha,\beta_0) \le ((b+j-2)/(D-2))^{\lfloor (b-2)/2\rfloor}$. The original proof bounded this ratio by 1.

(iii) arcsin *terminal cap* (Lemma 6). The terminal bound improves from $(8\epsilon/r)^{D-1}$ to $\frac{1}{2}(\sin\epsilon/r)^{D-1}$. This makes the argument uniform down to $\lambda = \log_2 10$ and removes the separate small-λ branch of [1].

The original method fails at a definite step. With the split fixed at $1/2$, every choice of parameters gives at most 0.5176, and at most 0.3336 with the original ratio estimates (Section 5.1). The moving split is therefore necessary, not a re-tuning. Section 5 records which change contributes which factor of the total $2^{-16} \to 1.3847$ (certified) improvement, a factor of about 9.07×10^4.

2 Three estimates

Write $f_{A,B}$ for the Beta(A,B) density and B for the beta function. Recall that $\mathrm{B}(A,B)$ decreases in each argument and that $\mathrm{B}(A,B+1) = \mathrm{B}(A,B)\,B/(A+B)$.

2.1 The terminal cap

Lemma 6 (arcsin cap)**.** *Let H be a Euclidean space of dimension $D \geq 2$, let $w \in H$ and $0 < \rho < 1$. Then*

$$\sigma_H\{u \in S(H) : \|u - w\| \leq \rho\} \;\leq\; \tfrac{1}{2}\, I_{\rho^2}\Big(\tfrac{D-1}{2}, \tfrac{1}{2}\Big) \;\leq\; \tfrac{1}{2}\rho^{D-1},$$

where $I_x(a, b)$ is the regularized incomplete beta function. Consequently, on an affine subsphere $z+rS(H)$ with $z \perp H$, $\|z\|^2+r^2 = 1$, the angular success probability $\sigma_H\{u : \arccos\langle \widehat{s}, z+ru\rangle \leq \epsilon\}$ of any fixed unit output $\widehat{s}$, or of any target-independent output law, is at most $\min\{1, \frac{1}{2}(\sin\epsilon/r)^{D-1}\} \leq (\sin\epsilon/r)^p$ for every $0 \leq p \leq D-1$.

Proof. If $\|u - w\| \leq \rho < 1 = \|u\|$, then $1 - 2\langle u, w\rangle + \|w\|^2 \leq \rho^2 < 1$, so $\langle u, w\rangle > \|w\|^2/2 \geq 0$ and $w \neq 0$. The distance from u to the line $\mathbb{R}w$ is at most $\|u - w\| \leq \rho$. Since $\langle u, w\rangle > 0$, the angle ϕ between u and $e = w/\|w\|$ is acute with $\sin\phi \leq \rho$, so $\phi \leq \theta := \arcsin\rho$. The cap $\{\phi \leq \theta\}$ with $\theta \leq \pi/2$ has mass $\frac{1}{2}\mathbb{P}(1 - \langle u, e\rangle^2 \leq \rho^2) = \frac{1}{2}I_{\rho^2}(\frac{D-1}{2}, \frac{1}{2})$, because $1 - \langle u, e\rangle^2 \sim \mathrm{Beta}(\frac{D-1}{2}, \frac{1}{2})$. With $a = (D-1)/2$ and the substitution $t = \rho^2 s$,

$$I_{\rho^2}(a, \tfrac{1}{2}) = \frac{\rho^{2a}}{\mathrm{B}(a, \frac{1}{2})}\int_0^1 s^{a-1}(1 - \rho^2 s)^{-1/2}ds \leq \frac{\rho^{2a}}{\mathrm{B}(a, \frac{1}{2})}\int_0^1 s^{a-1}(1 - s)^{-1/2}ds = \rho^{D-1}.$$

For the consequence: if $x \in S^{d-1}$ and $\angle(x, \widehat{s}) \leq \epsilon$, then $\|x - \cos\epsilon\,\widehat{s}\|^2 = 1 - 2\cos\epsilon\langle x, \widehat{s}\rangle + \cos^2\epsilon \leq \sin^2\epsilon$. Apply this to the unit vector $x = z+ru$ and project onto H (recall $z \perp H$): $\|u-w\| \leq \sin\epsilon/r$ with $w = \cos\epsilon\, P_H\widehat{s}/r$. If $\sin\epsilon/r < 1$, apply the first part with $\rho = \sin\epsilon/r$. Otherwise use total mass $1 \leq (\sin\epsilon/r)^p$. Averaging over an output law independent of the target preserves the bound. □

This replaces [1, Lemma 3.1, eq. (12) and its proof, `02-subspace-moments.tex`:24–56]. There the successful set was placed in a chordal ball of radius ϵ/r, recentred at a successful direction (doubling the radius), and then bounded by the angular cap $(2\theta)^{D-1}$ with $\theta \leq (\pi/2)\cdot 2\epsilon/r$. This gives $(2\pi\epsilon/r)^{D-1}$, rounded to $(8\epsilon/r)^{D-1}$. Using the centre $\cos\epsilon\,\widehat{s}$ instead of $\widehat{s}$ also replaces ϵ by $\sin\epsilon$, which makes the lower bound match the Kaczmarz bound exactly at the level of $d\ln(1/\sin^2\epsilon)$. The worst-case centre for a chordal ball is not on the sphere. It is at distance $\sqrt{1-\rho^2}$ from the origin, which is why the exact answer is the arcsin cap. This was checked by Monte Carlo (`computations/independent/indep_cap_kacz.py`, part f2).

2.2 The moving split

We use the setting of [1, Lemma 5.1]. Let E have dimension D', let $V \subset E$ have dimension $1 \leq j < n < D'$, let N be uniform among the n-dimensional subspaces of E containing V, and let v be uniform on $S(N)$ given N. Put $b = D' + 1 - n \geq 2$ and

$$\alpha = \tfrac{n-j}{2}, \quad \alpha_* = \tfrac{D'-j}{2} = \alpha + \tfrac{b-1}{2}, \quad \beta_0 = \tfrac{j}{2}, \quad \beta_1 = \tfrac{D'-n+j-1}{2} = \beta_0 + \tfrac{b-2}{2}.$$

Let $\nu_0 = \sigma_E$, and let ν_1 be the law of $\sqrt{1-Z}\,e_0 + \sqrt{Z}\,u_0$ with $Z \sim \mathrm{Beta}(\alpha, \alpha_* - \alpha)$, $e_0 \sim \sigma_V$, and $u_0 \sim \sigma_{E\cap e_0^\perp}$ given e_0, all independent.

Lemma 7 (Moving-split domination)**.** *For every $0 < \delta < 1$, as measures on $S(E)$,*

$$\mathrm{Law}(v) \;\leq\; A_\delta\,\nu_0 + B_\delta\,\nu_1, \qquad A_\delta = \frac{\mathrm{B}(\alpha_*, \beta_0)}{\mathrm{B}(\alpha, \beta_0)}(1-\delta)^{-\frac{b-1}{2}} \leq (1-\delta)^{-\frac{b-1}{2}}, \quad B_\delta = \frac{\mathrm{B}(\alpha, \beta_1)}{\mathrm{B}(\alpha, \beta_0)}\,\delta^{-\frac{b-2}{2}}.$$

Proof. Let $Q(y) = \|P_{V^\perp} y\|^2$. As in the proof of [1, Lemma 5.1] (`07-beta-mixture.tex`:50–67), $Q(v) \sim \mathrm{Beta}(\alpha, \beta_0)$, $Q \sim \mathrm{Beta}(\alpha_*, \beta_0)$ under ν_0, and $Q \sim \mathrm{Beta}(\alpha, \beta_1)$ under ν_1. The three laws are invariant under $O(V) \times O(V^\perp)$, and their two component lengths are a.s. nonzero. Hence a pointwise inequality between the scalar densities of Q lifts to the vector measures (*ibid.*, lines 69–79). Put $q_0 = 1 - \delta$. For $q \geq q_0$,

$$f_{\alpha,\beta_0}(q) = \frac{\mathrm{B}(\alpha_*, \beta_0)}{\mathrm{B}(\alpha, \beta_0)}\, q^{-\frac{b-1}{2}}\, f_{\alpha_*,\beta_0}(q) \leq A_\delta\, f_{\alpha_*,\beta_0}(q),$$

and $\mathrm{B}(\alpha_*, \beta_0) \leq \mathrm{B}(\alpha, \beta_0)$ since $\alpha_* > \alpha$. For $q < q_0$ we have $1 - q > \delta$, so

$$f_{\alpha,\beta_0}(q) = \frac{\mathrm{B}(\alpha, \beta_1)}{\mathrm{B}(\alpha, \beta_0)}(1-q)^{-\frac{b-2}{2}}\, f_{\alpha,\beta_1}(q) \leq B_\delta\, f_{\alpha,\beta_1}(q).$$

Hence $f_{\alpha,\beta_0} \leq A_\delta f_{\alpha_*,\beta_0} + B_\delta f_{\alpha,\beta_1}$ on $(0,1)$. □

At $\delta = 1/2$ and with $\mathrm{B}(\alpha, \beta_1) \leq \mathrm{B}(\alpha, \beta_0)$ this is the original $\mathrm{Law}(v) \leq 2^{(b-1)/2}(\nu_0 + \nu_1)$. The statement there is the weaker $2^{D'}(\nu_0 + \nu_1)$. Monotonicity of the beta function is the only input, and it holds for every q_0.

2.3 Beta ratio and inverse moment

Lemma 8 (Beta-ratio bound)**.** *For $\alpha, \beta_0 > 0$ and $h \geq 0$, $\dfrac{\mathrm{B}(\alpha, \beta_0 + h)}{\mathrm{B}(\alpha, \beta_0)} \leq \Big(\dfrac{\beta_0 + h}{\alpha + \beta_0 + h}\Big)^{\lfloor h \rfloor}$. In the notation above, $\alpha + \beta_1 = (D-2)/2$ with $D = D' + 1$, so*

$$\frac{\mathrm{B}(\alpha, \beta_1)}{\mathrm{B}(\alpha, \beta_0)} \leq \Big(\frac{b + j - 2}{D - 2}\Big)^{\lfloor (b-2)/2 \rfloor}.$$

Proof. $\mathrm{B}(\alpha, \beta_0 + h) \leq \mathrm{B}(\alpha, \beta_0 + \lfloor h \rfloor) = \mathrm{B}(\alpha, \beta_0) \prod_{i=0}^{\lfloor h \rfloor - 1} \frac{\beta_0 + i}{\alpha + \beta_0 + i}$. Each factor is at most $(\beta_0 + h)/(\alpha + \beta_0 + h)$, since $x \mapsto x/(\alpha + x)$ increases. For the second claim, $\alpha + \beta_1 = \frac{n-j}{2} + \frac{D' - n + j - 1}{2} = \frac{D'-1}{2}$ and $\beta_1 = \frac{b+j-2}{2}$. □

Lemma 9 (Inverse moment)**.** *If $Z \sim \mathrm{Beta}(\alpha, h_1)$ and $\ell < \alpha - 1$ is a nonnegative integer, then $\mathbb{E} Z^{-\ell} = \prod_{t=1}^{\ell} \frac{\alpha + h_1 - t}{\alpha - t} \leq \big(\frac{\alpha - 1}{\alpha - 1 - \ell}\big)^{h_1}$. (The tempting bound $(\alpha/(\alpha - \ell))^{h_1}$ is false for $h_1 = \frac{1}{2}$, e.g. $\alpha = \frac{5}{2}, \ell = 1$: $4/3 > \sqrt{5/3}$.)*

Proof. The product formula is [1, eq. (6)]. We have $\ln \mathbb{E} Z^{-\ell} = \sum_{t=1}^{\ell} \ln(1 + \frac{h_1}{\alpha - t}) \leq h_1 \sum_{t=1}^{\ell} \frac{1}{\alpha - t}$ and $\frac{1}{\alpha - t} \leq \int_t^{t+1} \frac{dx}{\alpha - x}$, since the integrand increases on $[t, t+1] \subset [0, \alpha - 1)$; summing over $1 \leq t \leq \ell$ gives $h_1 \ln \frac{\alpha - 1}{\alpha - 1 - \ell}$. □

Proposition 10 (One replica direction)**.** *In the setting of Lemma 7, let $\ell < \alpha - 1$ be an integer, and let $r_1 \geq (b + j - 2)/(D - 2)$ and $r_2 \geq (\alpha - 1)/(\alpha - 1 - \ell)$ with $r_1 < 1 \leq r_2$. If $r_1 r_2 \leq \delta < 1$, then*

$$A_\delta + B_\delta\, \mathbb{E} Z^{-\ell} \;\leq\; (1 - \delta)^{-\frac{b-1}{2}} + r_2 \delta^{-1/2}.$$

Proof. By Lemmas 8 and 9 with $h_1 = (b-1)/2$, $B_\delta \mathbb{E} Z^{-\ell} \leq r_1^{\lfloor (b-2)/2 \rfloor} \delta^{-(b-2)/2} r_2^{(b-1)/2}$. For even b this equals $(r_1 r_2/\delta)^{(b-2)/2} r_2^{1/2} \leq r_2^{1/2}$. For odd $b \geq 3$ it equals $(r_1 r_2/\delta)^{(b-3)/2} \delta^{-1/2} r_2 \leq r_2 \delta^{-1/2}$. □

The point is the scaling. For a block of $b \leq k = \kappa d$ rows, the first term costs $\frac{\kappa}{2} \ln \frac{1}{1-\delta}$ nats per unit dimension. The second term is $O(1)$ per direction, hence $O(1/d)$ per unit dimension, provided δ exceeds $r_1 r_2 \approx (\kappa + \mu)/(1 - \pi/(1 - \kappa - \mu))$. Since this threshold for δ tends to 0 with κ, μ, the first term is $o(\kappa)$ as $\kappa, \mu \to 0$, while the main residual cost is of order $\kappa\pi$.

3 Proof of Theorem 2

Rate function. For parameters $\kappa, \pi, \mu, \zeta > 0$ with $\kappa + \mu + \pi < 1$ put

$$\delta = \frac{(1+\zeta)(\kappa+\mu)}{1-\pi/(1-\kappa-\mu)}, \qquad R_1(\kappa,\pi) = \int_0^{\pi/2} \ln \frac{1/2 - x}{(1-\kappa)/2 - x}\, dx,$$

$$g = R_1(\kappa,\pi) + \frac{\kappa}{2} \ln \frac{1}{1-\delta}, \qquad \Phi(\kappa,\pi,\mu,\zeta) = \frac{\kappa\pi \ln 2}{g} - \frac{\kappa}{\log_2 10},$$

and call the parameters admissible if moreover $\delta < 1$.

Lemma 11. $\sup\{\Phi : \textit{admissible}\} = 2\ln 2$. *Explicitly,* $\Phi(10^{-7}, 10^{-4}, 10^{-8}, 10^{-3}) \geq 1.38470$. *This was certified with 200-bit interval arithmetic (`computations/certify.py`) and recomputed independently with `mpmath` quadrature (`computations/independent/indep_constant.py`).*

Proof. Using $\ln(1+y) \leq y$, $R_1 \leq \frac{\kappa}{2} \int_0^{\pi/2} \frac{dx}{(1-\kappa)/2 - x} = \frac{\kappa}{2} \ln \frac{1-\kappa}{1-\kappa-\pi}$. Take $\zeta = 1$, $\mu = \kappa^2$, and fix $\pi \in (0, \frac{1}{2})$. As $\kappa \to 0$ we have $\delta = O(\kappa)$, so $g \leq \frac{\kappa}{2}[\ln \frac{1}{1-\pi} + O(\kappa)]$ and $\Phi \geq \frac{2\pi \ln 2}{\ln(1/(1-\pi))} - O(\kappa)$. The right side tends to $2\ln 2$ as $\pi \to 0$. Conversely, $R_1 \geq \frac{\kappa}{2} \ln \frac{1}{1-\pi}$ gives $\Phi < 2\pi \ln 2 / \ln \frac{1}{1-\pi} < 2 \ln 2$. □

Setup. Write $\lambda_s = \log_2(1/\sin\epsilon)$. Then $\lambda \leq \lambda_s \leq \lambda + 0.0025$ and $\lambda_s \geq \log_2 10$ for $\epsilon \leq 1/10$. Suppose the theorem fails for some $c < 2\ln 2$. Then there is an unbounded sequence of dimensions d, accuracies $\epsilon_d \leq 1/10$, and learners with memory $M(d)$ and success $\geq 2/3$ such that $T < cd\lambda_s \leq 2cd\lambda$. Fix admissible $(\kappa, \pi, \mu, \zeta)$ with $\Phi > c$ (Lemma 11). By the short-run lemma [1, Lemma 4.2], applied with $2c$ and $\alpha = 2/3$, we have $\lambda = o(d)$, $T = o(d^2)$, and $\log_2 w = o(d^2)$ for the padded width $w = (T+2)2^M$. These bounds are uniform in the accuracy and the rules, because [1, eq. (18)] bounds $(d-2)\lambda$ by a function of d and $M(d)$. Set

$$k = \lfloor \kappa d \rfloor, \quad m = \lceil \mu d \rceil, \quad \ell = \lfloor \pi d/2 \rfloor, \quad p = 2\ell, \quad J = \lceil T/k \rceil \leq 2c\lambda_s/\kappa + 1 = o(d), \quad D_{\min} = d - 2J.$$

Put $\alpha_{\min} = (D_{\min} - k - m)/2$, $r_1 = (k+m)/(D_{\min} - 2)$ and $r_2 = (\alpha_{\min} - 1)/(\alpha_{\min} - 1 - \ell)$. Since $J = o(d)$, $r_1 \to \kappa + \mu$ and $r_2 \to (1 - \pi/(1-\kappa-\mu))^{-1}$, so $r_1 r_2 \to \delta/(1+\zeta) < \delta$. Hence for all large d in the sequence:

$$r_1 r_2 \leq \delta, \qquad p \leq D_{\min} - 1, \qquad m < D_{\min} - k, \qquad \ell + 1 < \alpha_{\min}. \tag{1}$$

Define

$$P_{\mathrm{res}} = \prod_{t=1}^{\ell} \frac{D_{\min}/2 - t}{(D_{\min} - k)/2 - t}, \qquad G = w^{1/m}\left[(1-\delta)^{-k/2} + r_2 \delta^{-1/2}\right] P_{\mathrm{res}}.$$

Invariant. Fix a seed as in [1, Lemma 2.1], pad to horizon T [1, Lemma 4.1], and cut the horizon into J batches of length $\leq k$. For a state v after i batches let $h_{i,v}(s)$ be its suffix success. We claim that every affine subsphere $c_0 + rS(H)$ with $\dim H = D \geq d - 2i$ satisfies

$$\int h_{i,v}(c_0 + ru)\, d\sigma_H(u) \leq G^{J-i} (\sin\epsilon / r)^p. \tag{2}$$

At $i = J$ this is Lemma 6, since $D \geq D_{\min} \geq p + 1$.

One block. Assume (2) at $i+1$. Let the batch have length $1 \le b \le k$ and let $D \ge d-2i$. By [1, Lemmas 2.4 and 2.5], $\rho^2 \sim \mathrm{Beta}((D-b)/2, b/2)$, $r' = r\rho$, $E = H \cap e^\perp$, $D' = D-1$, $n = D-b$. The left side of (2) is at most $\mathbb{E}_{\rho,e}\mathbb{E}_N \max_{v'} F_{v'}(N)$, where $F_{v'}(N) = \int h_{i+1,v'}(c' + r'u)\, d\sigma_N(u)$. Put $U_0 = G^{J-i-1}(\sin\epsilon/r')^p$. Expand $F_{v'}(N)^m$ over m conditionally independent directions. For $j = 0$ or $b = 1$ the next direction is uniform on $S(E)$ [1, Lemma 3.2], and its factor is $\le U_0$. For $j \ge 1$ and $b \ge 2$, apply Lemma 7. Under ν_0 the factor is $\le U_0$, by the stage-$(i+1)$ bound in dimension $D-1$. Under ν_1, given (Z, e_0), the target is uniform on an admissible affine subsphere of dimension $D-2 \ge d-2(i+1)$ and radius $r'\sqrt{Z}$ ([1], `07-beta-mixture.tex`:224–237), so the factor is $\le U_0 Z^{-\ell}$. Now $\alpha = (D-b-j)/2 > \alpha_{\min}$ and $(b+j-2)/(D-2) \le r_1$, so by Proposition 10 and (1) the conditional factor is at most $[(1-\delta)^{-k/2} + r_2\delta^{-1/2}]\, U_0$. Integrating in reverse conditional order, applying Jensen's inequality and summing over at most w states [1, Lemma 2.5] gives $\mathbb{E}_N \max_{v'} F_{v'} \le w^{1/m}[(1-\delta)^{-k/2} + r_2\delta^{-1/2}]\, U_0$. Finally U_0 contains ρ^{-p}, and $\mathbb{E}\rho^{-p} = \prod_{t=1}^{\ell} \frac{D/2-t}{(D-b)/2-t} \le P_{\mathrm{res}}$. Each factor increases in b and decreases in $D \ge D_{\min}$. This proves (2) at i.

Conclusion. At $i = 0$ (with $H = \mathbb{R}^d$, $r = 1$), averaging over the seed and the initial state as in [1] gives $2/3 \le G^J(\sin\epsilon)^p = G^J 2^{-p\lambda_s}$. Since $J \le T/k + 1$,

$$T \;\ge\; k\Big(\frac{p\lambda_s \ln 2 - \ln(3/2)}{\ln G} - 1\Big). \tag{3}$$

We have $\ln G \le \frac{\ln w}{m} + \frac{k}{2}\ln\frac{1}{1-\delta} + \ln(1 + r_2\delta^{-1/2}) + \ln P_{\mathrm{res}}$. Here $\ln w/(md) \to 0$, because $M = o(d^2)$ and $m \ge \mu d$; this is the only place where $o(d^2)$ memory enters besides the short-run lemma. The factors of P_{res} increase in t, so $\ln P_{\mathrm{res}} \le \int_0^{\ell+1} \ln\frac{D_{\min}/2-x}{(D_{\min}-k)/2-x}dx = (1+o(1))\, dR_1(\kappa,\pi)$, because $D_{\min}/d \to 1$. Thus $\limsup \ln G/d \le g$. Also $\ln G \ge \ln P_{\mathrm{res}} \ge \ell \ln(1+k/d)$, which grows linearly in d. Dividing (3) by $d\lambda_s$ and using $\lambda_s \ge \log_2 10$ gives $\liminf T/(d\lambda_s) \ge \kappa\pi \ln 2/g - \kappa/\log_2 10 = \Phi > c$. This contradicts $T < cd\lambda_s$. Every eventual condition depends only on d, $M(d)$ and the fixed parameters, which gives the threshold $d_0(c, M)$. □

Remark 12. The small-accuracy branch of [1] ($\lambda \le 1024$, handled by the full-data bound $T > d/4$) is no longer needed. With terminal constant 1, the induction itself is effective for all $\lambda_s \ge \log_2 10$. The $-\kappa/\log_2 10$ term in Φ is the price of this uniformity, and it vanishes as $\kappa \to 0$.

4 Finite-bit Kaczmarz: proof of Lemma 3

The learner. Let $\mathcal{G}_q = 2^{-q}\mathbb{Z} \cap [-3, 3]$. Then $|\mathcal{G}_q| = 6 \cdot 2^q + 1 \le 2^{q+3}$, and the state space is $\mathcal{G}_q^d$, so $M = d(q+3)$ bits suffice. Start with $z_0 = 0$. On reading the real pair (x_t, y_t) with $x_t \ne 0$, compute in exact arithmetic

$$\tilde{z}_t = z_{t-1} + \frac{y_t - \langle x_t, z_{t-1}\rangle}{\|x_t\|^2} x_t,$$

and store $z_t = \mathrm{rd}_q(\tilde{z}_t)$, where rd_q rounds each coordinate to the nearest point of $2^{-q}\mathbb{Z}$ (ties toward 0; a Borel rule). On the null event $x_t = 0$ keep $z_t = z_{t-1}$. After T samples output $z_T/\|z_T\|$, or a fixed unit vector if $z_T = 0$. This is a learner of Definition 1: the real sample is used only inside the transition, and only z_t is retained. We show below that $\tilde{z}_t \in [-3, 3]^d$, so $z_t \in \mathcal{G}_q^d$.

Error recursion. Fix $s \in S^{d-1}$ and put $u_t = x_t/\|x_t\|$, $P_t = I - u_t u_t^{\mathsf{T}}$, $e_t = z_t - s$. Since $y_t = \langle x_t, s\rangle$ exactly, $\tilde{z}_t - s = P_t e_{t-1}$. Hence

$$e_t = P_t e_{t-1} + r_t, \qquad \|r_t\|_\infty \le 2^{-q-1}, \qquad \|r_t\|_2 \le \delta_q := \sqrt{d}\, 2^{-q-1}.$$

Let $\hat{e}_t = P_t \cdots P_1(-s)$ be the error of exact Kaczmarz on the same rows, and let $a_t = e_t - \hat{e}_t$. Then $a_t = P_t a_{t-1} + r_t$, $a_0 = 0$, so $a_T = \sum_{t=1}^{T} P_T \cdots P_{t+1} r_t$. As $\|P_t\| \le 1$, we get $\|a_t\| \le t\delta_q$ deterministically.

Range. We have $\|e_{t-1}\| \le \|\hat{e}_{t-1}\| + \|a_{t-1}\| \le 1 + T\delta_q$, and so $\|\tilde{z}_t\|_\infty \le \|s\| + \|P_t e_{t-1}\| \le 2 + T\delta_q$. We claim $T\delta_q \le 1$. Indeed $T \le (1+\eta)d\Lambda + 1$ with $\Lambda = \ln(1/\sin^2\epsilon) = 2\ln 2 \cdot \lambda'$ and $\lambda' = \log_2(1/\sin\epsilon)$. Hence $\log_2(T\sqrt{d}/2) \le \frac{3}{2}\log_2 d + \log_2(2.78\lambda' + 1) - 1$. Since $2^{\lambda'+4.29} \ge 19.5(1 + 0.69\lambda') \ge 2.78\lambda' + 1$, this is $\le \frac{3}{2}\log_2 d + \lambda' + 3.29 \le q$. So $\tilde{z}_t \in [-3,3]^d$, and rounding to the nearest grid point stays in $[-3,3]$ because $\pm 3 \in \mathcal{G}_q$.

Exact part: a Beta product. Let $\mathcal{F}_t = \sigma(x_1, \ldots, x_t)$. If $\hat{e}_{t-1} \ne 0$, put $B_t = \langle u_t, \hat{e}_{t-1}/\|\hat{e}_{t-1}\|\rangle^2$. Otherwise put $B_t = \langle u_t, e_1\rangle^2$. Since u_t is uniform and independent of $\mathcal{F}_{t-1}$, $B_t \sim \mathrm{Beta}(\frac{1}{2}, \frac{d-1}{2})$ independently of $\mathcal{F}_{t-1}$. Hence the B_t are i.i.d., with $\mathbb{E}B_t = 1/d$ and $\operatorname{Var} B_t = \frac{2(d-1)}{d^2(d+2)} \le \frac{2}{d^2}$. Also $\|\hat{e}_T\|^2 = \prod_t (1 - B_t) \le \exp(-\sum_t B_t)$. Let $\theta = \eta/4$ and $L = \Lambda + 2\ln\frac{1}{1-\theta}$. For $\epsilon \le 1/10$ we have $\Lambda \ge \ln(1/\sin^2 0.1) > 4.6$, and $2\ln\frac{1}{1-\theta} \le \frac{8}{3}\theta = \frac{2}{3}\eta$, so $L \le \Lambda(1+\eta/6)$. With $1+\gamma = (1+\eta)/(1+\eta/6)$, so that $\gamma \ge 5\eta/7$, we get $T/d \ge (1+\gamma)L$. Chebyshev's inequality gives

$$\mathbb{P}\Big(\sum_{t \le T} B_t < L\Big) \le \frac{2T/d^2}{(T/d)^2\gamma^2/(1+\gamma)^2} = \frac{2(1+1/\gamma)^2}{T} \le \frac{11.52}{\eta^2 T} \le \frac{2.51}{\eta^2 d} \le \frac{1}{6},$$

using $(1 + 1/\gamma) \le 2.4/\eta$, $T \ge 4.6d$ and $d \ge 16/\eta^2$. Off this event, $\|\hat{e}_T\| \le e^{-L/2} = (1-\theta)\sin\epsilon$.

Rounding part: contraction in L^2. The vector r_t is $\mathcal{F}_t$-measurable, and $u_{t+1}, \ldots, u_T$ are i.i.d. uniform and independent of $\mathcal{F}_t$. For a fixed w we have $\mathbb{E}\|P_u w\|^2 = (1 - 1/d)\|w\|^2$. Iterating, $\mathbb{E}[\|P_T \cdots P_{t+1} r_t\|^2 \mid \mathcal{F}_t] = (1-1/d)^{T-t}\|r_t\|^2$. By Minkowski's inequality in $L^2(\mathbb{P})$,

$$\big(\mathbb{E}\|a_T\|^2\big)^{1/2} \le \sum_{t=1}^{T} (1-1/d)^{(T-t)/2}\delta_q \le \frac{\delta_q}{1 - \sqrt{1-1/d}} \le 2d\delta_q = d^{3/2}2^{-q}.$$

By the choice of q, $2^q \ge 4\sqrt{6}\, d^{3/2}/(\eta \sin\epsilon)$, so $(\mathbb{E}\|a_T\|^2)^{1/2} \le \theta \sin\epsilon/\sqrt{6}$. Markov's inequality on $\|a_T\|^2$ then gives $\mathbb{P}(\|a_T\| > \theta\sin\epsilon) \le 1/6$.

Conclusion. With probability at least $1 - \frac{1}{6} - \frac{1}{6} = \frac{2}{3}$, $\|z_T - s\| \le \|\hat{e}_T\| + \|a_T\| \le \sin\epsilon < 1$, so $z_T \ne 0$. The point z_T lies in the closed ball of radius $\sin\epsilon$ about the unit vector s, and every ray from the origin meeting this ball makes an angle at most $\arcsin(\sin\epsilon) = \epsilon$ with s. Hence $\arccos\langle z_T/\|z_T\|, s\rangle \le \epsilon$. For the bit count, $\log_2(1/\sin\epsilon) \le \lambda + 0.0025$ for $\epsilon \le 1/10$ (as $\sin\epsilon \ge \epsilon(1-\epsilon^2/6)$). Then $q + 3 \le \lambda + \frac{3}{2}\log_2 d + \log_2\frac{1}{\eta} + \log_2(4\sqrt{6}) + 4.0025 \le \lambda + \frac{3}{2}\log_2 d + \log_2\frac{1}{\eta} + 7.3$. Finally, $0 \le \ln\frac{1}{\sin^2\epsilon} - \ln\frac{1}{\epsilon^2} = 2\ln\frac{\epsilon}{\sin\epsilon} \le -2\ln(1-\epsilon^2/6) \le \epsilon^2/2$, and $\frac{\epsilon^2/2}{\ln(1/\epsilon^2)} \le \frac{0.005}{4.605} < 0.0011$. □

Remark 13 (Where the bits go). Rounding cost λ bits plus $\frac{3}{2}\log_2 d + O(1)$ bits per coordinate: $\frac{1}{2}\log_2 d$ for $\ell_\infty \to \ell_2$, and $\log_2 d$ for the effective memory $\sum_k (1-1/d)^{k/2} \le 2d$ of the contraction. The L^2 argument avoids a $\log_2 T \approx \log_2(d\lambda)$ term, which is why C does not depend on ϵ. We did not try to optimize the $\frac{3}{2}$. The λ term cannot be avoided by any learner: the final output takes at most 2^M values (for each fixed sample sequence), each successful only for $\hat{s}$ in a cap of angular radius ϵ, whose normalized measure is at most $\frac{1}{2}(\sin\epsilon)^{d-1}$ by the cap bound used in the terminal lemma; averaging over a uniform $\hat{s}$, success probability $\ge 2/3$ forces $2^M \cdot \frac{1}{2}(\sin\epsilon)^{d-1} \ge \frac{2}{3}$, i.e. $M \ge (d-1)\lambda' + \log_2\frac{4}{3}$. (The last column of Table 3, with $\lceil\lambda'\rceil - 3$ fractional bits and no $\frac{3}{2}\log_2 d$ term, only illustrates that the algorithm fails below this count.)

Remark 14 (Probability estimate). Chebyshev's inequality is enough because $\operatorname{Var}\sum B_t = O(T/d^2)$ while the required deviation is $\eta T/d$. Bernstein's inequality for the bounded variables $B_t \in [0,1]$, or the exact formulas $\mathbb{E}\ln\frac{1}{1-B} = \psi(\frac{d}{2}) - \psi(\frac{d-1}{2})$ and $\operatorname{Var}\ln\frac{1}{1-B} = \psi'(\frac{d-1}{2}) - \psi'(\frac{d}{2})$, would give exponential tails and a threshold $d_0 = O(\eta^{-2})$ with a better constant. This is not needed for the first-order statement.

5 Where the constant was lost

Table 1 evaluates the lower bound (3), i.e. $T_{\mathrm{LB}}/(d\lambda)$, after each successive correction. Rows S1–S5 use the paper's parameters $k = m = \lfloor d/16 \rfloor$, $\ell = \lfloor d/32 \rfloor$ at $d = 4 \cdot 10^4$, with exact beta products (script `computations/ledger.py`). Row S6 is the parameter limit (Lemma 11).

step	change	c ($\lambda \to \infty$)	factor
S0	paper as printed, `07-beta-mixture.tex`:117	$1.53 \times 10^{-5} = 2^{-16}$	–
S1	same displayed bounds, no slack ($k, p \geq d/16$ not $d/32$; $\log_2 G = 4d$ not $5d$; $\lambda - 3 \approx \lambda$)	$9.77 \times 10^{-4} = 2^{-10}$	$\times 64$
S2	exact values of the stated lemma bounds $2^{D'}(1 + \mathbb{E}Z^{-\ell})$, $\mathbb{E}\rho^{-p}$	3.88×10^{-3}	$\times 4.0$
S3	constant the proof actually yields, $2^{(b-1)/2}(1 + \mathbb{E}Z^{-\ell})$	0.1048	$\times 27$
S3b	split at 1/2, exact beta ratios	0.1195	$\times 1.14$
S4	moving split $q_0 = 1 - \delta$, optimized per j (Lemma 7)	0.8704	$\times 7.3$
S5	terminal cap $8 \to 1$ (Lemma 6); at $\lambda = \log_2 10$: $0.0655 \to 0.8515$	0.8704	($\times 13$ at $\lambda = \log_2 10$)
S6	parameters $\mu \ll \kappa \ll \pi \to 0$ (Lemma 11)	$\to 2\ln 2 = 1.3863$	$\times 1.59$
	certified uniform constant at $(10^{-7}, 10^{-4}, 10^{-8})$	1.38470	total $\times 9.07 \cdot 10^4$

Table 1: Ledger of the constant. The largest single losses are the arithmetic slack (S1, $\times 64$), the statement-versus-proof gap of the mixture lemma (S3, $\times 27$), and the fixed split (S4, $\times 7.3$). The terminal constant matters only for bounded λ, where it decides whether the bound is uniform at all.

5.1 Limit of the original framework

Keep the original split $q_0 = 1/2$ but free all parameters (κ, π, μ) and use terminal constant 1. The first component of the mixture then costs $\frac{b-1}{2}\ln 2$ per direction, which is $\frac{\kappa}{2}\ln 2$ nats per unit dimension per block. This holds whatever the parameters are. With the paper's ratio estimate $\mathrm{B}(\alpha, \beta_1) \leq \mathrm{B}(\alpha, \beta_0)$, the second component additionally costs $\mathbb{E}Z^{-\ell} = e^{(\kappa/2)d\ln\frac{1}{1-\pi}(1+o(1))}$. As $\kappa, \mu \to 0$ the rate becomes $g_{1/2} = \frac{\kappa}{2}[\ln 2 + 2\ln\frac{1}{1-\pi}]$, and

$$c_{1/2} = \sup_{0<\pi<1} \frac{2\pi \ln 2}{\ln 2 + 2\ln\frac{1}{1-\pi}} = \frac{\ln 2}{u} = 0.33357\ldots, \qquad u = -W_{-1}\Big(-\frac{1}{e\sqrt{2}}\Big) = 2.07796\ldots,$$

attained at $\pi^* = 1 - 1/u = 0.51876$. Here W_{-1} is the lower branch of the Lambert function. The first-order condition $\ln 2 + 2\ln u = 2(u-1)$, with $u = 1/(1-\pi)$, gives $c_{1/2} = \ln 2\,(1 - \pi^*)$. Numerical optimization over all (κ, π, μ) in the original framework confirms 0.333571 (`computations/rates.py`, mode `half`), which is 4.16 times below $2\ln 2$. With the exact beta ratio but still $q_0 = 1/2$, the second component becomes $O(1)$, but the first does not. This gives $\sup_\pi 2\pi\ln 2/\ln\frac{2}{1-\pi} = 2\ln 2/u' = 0.51759$ with $u' = -W_{-1}(-1/(2e)) = 2.67835$. Either way the cap is a fixed fraction of $2\ln 2$, so the split point must move to 1.

6 Finite d

The convergence in Theorem 2 is slow. The constant approaches $2\ln 2$ only when $\mu \ll \kappa \ll \pi \to 0$. The integrality requirements $k, m \geq 2$ alone need $d \geq 2/\mu$. The memory term $\ln w/m$ must also be small compared with $d\,g \approx \frac{\kappa}{2}\pi d$, i.e. $M/d^2 \ll \mu\kappa\pi$. Table 2 reports the effective constant $T_{\mathrm{LB}}/(d\lambda)$ from (3) with exact beta products, worst test dimension $D_{\min}$, $\lambda = 10$, and Kaczmarz memory $M = d(\lambda + 2\log_2 d)$, maximized over a parameter grid. These are estimates and not certificates: the block maximum is evaluated at $b \in \{k-1, k\}$ and $j \in \{1, m/2, m-1\}$.

d	10^4	10^6	10^8	10^{10}
best value found (estimate, not a certificate)	0.341	0.779	1.071	1.231

Table 2: Effective lower-bound constant at finite d, $\lambda = 10$, memory $d(\lambda + 2\log_2 d)$ (compare $2\ln 2 = 1.386$ and the original 2^{-16}). Each entry is the maximum over the union of two parameter grids (`computations/independent/indep_constant.py`, $\kappa \leq 0.03, \pi \leq 0.2$; `computations/finite_d_fine.py`, $\kappa \leq 0.3, \pi \leq 0.7$); the 10^8 entry comes from the first grid, the others from the second. Optimizers: $(0.076, 0.48, 0.13)$, $(0.038, 0.33, 0.024)$, $(0.01, 0.2, 0.01)$, $(0.0024, 0.10, 0.0019)$. Since the block maximum is evaluated only at $b \in \{k-1, k\}$, $j \in \{1, m/2, m-1\}$, these numbers are estimates of what the proof gives, not proved bounds.

So at $d = 10^4$ the improved proof gives about 0.34 (about $2.2 \cdot 10^4$ times the original 2^{-16}), which is still a factor 4 below Kaczmarz's 1.386. Even at $d = 10^{10}$ the gap is 11%. The statement "Kaczmarz is first-order optimal" is asymptotic. The rate at which the effective constant approaches $2\ln 2$ is not optimized here.

7 Numerical unit test of Lemma 3

Script `computations/kacz_fixedpoint.py` runs exactly the learner of Section 4, with float64 standing in for exact arithmetic. The rounding resolution is $2^{-q} \geq 2^{-35}$, far above the float64 error. It uses 200 independent signals per row and $\eta = 1/4$. The lemma's sufficient condition $d \geq 16/\eta^2 = 256$ fails at $d = 200$. Those rows test the algorithm, not the lemma's Chebyshev constant.

d	ϵ	q	M/d	T	succ. fix/real	rms$\|a_T\|$	bound	$\hat{\mathbb{P}}(\sum B < L)$	max$\|\tilde{z}\|_\infty$	1st hit	succ. q_{low}
200	0.1	21	24	1153	1.000/1.000	$2.7 \cdot 10^{-5}$	$1.3 \cdot 10^{-3}$	0.000	0.310	1.0307	0.000 ($q = 1$)
200	0.001	27	30	3454	1.000/1.000	$4.3 \cdot 10^{-7}$	$2.1 \cdot 10^{-5}$	0.000	0.312	0.9953	0.000 ($q = 7$)
200	1e-05	34	37	5757	1.000/1.000	$3.4 \cdot 10^{-9}$	$1.6 \cdot 10^{-7}$	0.000	0.303	0.9989	0.000 ($q = 14$)
400	0.1	22	25	2305	1.000/1.000	$2.8 \cdot 10^{-5}$	$1.9 \cdot 10^{-3}$	0.000	0.206	1.0036	0.000 ($q = 1$)
400	0.001	29	32	6908	1.000/1.000	$2.2 \cdot 10^{-7}$	$1.5 \cdot 10^{-5}$	0.000	0.216	1.0043	0.000 ($q = 7$)
400	0.0001	32	35	9211	1.000/1.000	$2.7 \cdot 10^{-8}$	$1.9 \cdot 10^{-6}$	0.000	0.236	1.0043	0.000 ($q = 11$)
1000	0.1	24	27	5761	1.000/1.000	$1.7 \cdot 10^{-5}$	$1.9 \cdot 10^{-3}$	0.000	0.150	0.9982	0.000 ($q = 1$)
1000	0.001	31	34	17270	1.000/1.000	$1.3 \cdot 10^{-7}$	$1.5 \cdot 10^{-5}$	0.000	0.152	1.0025	0.000 ($q = 7$)

Table 3: Fixed-point Kaczmarz at the lemma's T and q. Columns: success rate (fixed point / real registers); root-mean-square rounding accumulation $\|a_T\|$ versus the proved L^2 bound $d^{3/2}2^{-q}$; empirical $\mathbb{P}(\sum B_t < L)$; max$\|\tilde{z}_t\|_\infty$ (the proof needs ≤ 3); median first time with angular error $\leq \epsilon$, divided by $d\ln(1/\sin^2\epsilon)$ (checked every 50 steps, so the resolution is $50/(d\ln(1/\sin^2\epsilon)) \leq 4\%$); success with only $\lceil\lambda'\rceil - 3$ fractional bits.

All rows succeed with frequency $1.000 \geq 2/3$. The rounding term stays 47–115 times below

its proved bound, and $\|\tilde{z}_t\|_\infty \le 0.32$. The first hitting time is $(1.00 \pm 0.03)\, d \ln(1/\sin^2 \epsilon)$, which matches the first-order constant. With $\lambda - 3$ fractional bits the algorithm never succeeds. The unit test does not probe the optimality of the $\frac{3}{2} \log_2 d$ term.

A Line-by-line changes to the original proof

File references are to the LaTeX sources of [1] (`preprints/Subsphere-methods-.../build/`).

C1. `main.tex`:13–14 and `sections/00-introduction.tex`:53–67, 85–88: replace 2^{-16} by "any $c < 2 \ln 2$, with threshold $d_0(c, M)$".

C2. `02-subspace-moments.tex`:24–30, 44–56 (eq. `s:affine-terminal`): replace $\min\{1, (8\epsilon/R)^{D-1}\}$ by $\min\{1, \frac{1}{2}(\sin \epsilon/R)^{D-1}\}$, using Lemma 6: centre the chordal ball at $\cos \epsilon\, \widehat{s}$ (radius $\sin \epsilon$), do not recentre at a successful direction, and use the arcsin cap.

C3. `07-beta-mixture.tex`:21–48 (Lemma `s:mixture`): replace $\mathrm{Law}(v) \le 2^{D'}(\nu_0 + \nu_1)$ by Lemma 7 with a free $\delta \in (0, 1)$. Proof lines 86–106: split at $q_0 = 1 - \delta$ and keep the beta ratios.

C4. `07-beta-mixture.tex`:115–118: $c_* = 2^{-16}$ becomes the given $c < 2 \ln 2$. Choose admissible $(\kappa, \pi, \mu, \zeta)$ with $\Phi > c$ (Lemma 11).

C5. `07-beta-mixture.tex`:121–132: keep the short-run reduction with this c. Delete the branch $\lambda \le 1024$ (lines 130–132) and with it the use of `03-stopping.tex`:123–135.

C6. `07-beta-mixture.tex`:134–141: parameters $k = \lfloor \kappa d \rfloor$, $m = \lceil \mu d \rceil$, $\ell = \lfloor \pi d/2 \rfloor$, and $G = w^{1/m}[(1-\delta)^{-k/2} + r_2 \delta^{-1/2}] P_{\mathrm{res}}$ instead of $2^{4d} w^{1/m}$.

C7. `07-beta-mixture.tex`:149–153: budget $J \le 2c\lambda_s/\kappa + 1 = o(d)$, and the eventual conditions (1) replace $J \le d/8$, $m, p \ge d/32$.

C8. `07-beta-mixture.tex`:166–168: invariant $(8\epsilon/r)^p \to (\sin \epsilon/r)^p$. Lines 176–179: base case via Lemma 6.

C9. `07-beta-mixture.tex`:216: replace $D \ge 3d/4$, $n \ge 11d/16$ by $D \ge D_{\min}$, $n - j \ge D_{\min} - k - m$.

C10. `07-beta-mixture.tex`:239–257: replace eq. `(s:mixture-inverse)` ($\mathbb{E} Z^{-\ell} \le 2^d$) and $2^d U_0(1 + \mathbb{E} Z^{-\ell}) \le 2^{3d} U_0$ by Lemma 9 and Proposition 10: conditional factor $\le [(1-\delta)^{-k/2} + r_2 \delta^{-1/2}] U_0$.

C11. `07-beta-mixture.tex`:262–269: $(2^{3d} U_0)^m \to ([(1-\delta)^{-k/2} + r_2 \delta^{-1/2}] U_0)^m$.

C12. `07-beta-mixture.tex`:271–281: residual moment $2^\ell \le 2^d$ replaced by $P_{\mathrm{res}} = e^{(R_1(\kappa,\pi) + o(1))d}$. This uses monotonicity in b and D.

C13. `07-beta-mixture.tex`:283–299: replace the arithmetic `(s:numeric-arithmetic)` by (3) and the limit argument of Section 3.

C14. `07-beta-mixture.tex`:296–298 (side remark): the displayed chain uses $p \ge d/32$ and $J \le 32 c_* \lambda + 1$. Both are looser than needed ($p \approx d/16$, $k \approx d/16$). This is row S1 of Table 1, and it is not an error.

No step of the original proof was found to be wrong. All changes are sharpenings. The measurability, padding, Borel-version and short-run arguments (`01-residual-spheres.tex`, `03-stopping.tex`:13–88) are used verbatim.

B Reproducibility

Scripts (Python 3, `numpy`, `scipy`, `mpmath`) are in the directory `computations/` of the accompanying repository; `computations/run_all.sh` runs all of them: `ledger.py` (Table 1), `rates.py` (rate functions; framework limit 0.333571), `certify.py` (interval certificate 1.38470), `checks.py` and `independent/indep_mixture.py` (Monte Carlo of the beta laws, pointwise domination of Lemma 7 on grids including $q_0 \to 1$), `inverse_moment_check.py` (Lemma 9, including the counterexample to the tempting bound), `independent/indep_cap_kacz.py` (Lemma 6 for $D = 2, \dots, 399$), `independent/indep_constant.py`, `finite_d_fine.py` (Table 2), `finite_d_fullscan.py` (full scan over all (b, j) at the optimizers for $d = 10^4, 10^6$; it gives the same block maximum as the sampled points), `memratio.py` ($\Theta(d^2)$ remark), and `kacz_fixedpoint.py` (Table 3). The scripts in `computations/independent/` are separate re-implementations that do not import the others.

Acknowledgement and provenance

The results in this note were found and checked by Claude (Anthropic) agents working under the direction of [AUTHOR NAME]. The note has had no human peer review, and nothing in it is formalized in a proof assistant. The underlying architecture and all unchanged lemmas are from the OpenAI preprint [1].

References

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